

Abnik Ahilasamy

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Education

- Arizona State University — MS in Data Science, GPA 3.97/4.0** **May 2026, Tempe, AZ**
Coursework: Statistical Machine Learning, Data Processing at Scale, Computer Vision, Data Mining
- SRM Institute of Science and Technology — B.Tech. in AI & ML (Top 5%), GPA 3.8/4** **Jun 2024, Chennai, IN**
Coursework: Deep Learning, Artificial Intelligence, Neural Networks, NLP, Nature-Inspired Computing, Algorithms

Experience

- Larsen & Toubro — AI / LLM Engineer** **Apr 2024 – Jun 2024, Chennai, IN**
- Built **0 to 1 agentic RAG system** (LangChain, LlamaIndex), reducing hallucinations by **35%**
 - Owned the **end-to-end pipeline** (retrieval, reasoning, APIs, evaluation, infra), delivering a production-ready system
 - Re-architected inference and retrieval flow (caching, batching, memory optimization), achieving **40% lower latency** on workloads
 - Designed **evaluation framework** (faithfulness, groundedness, nDCG@10, MRR) for rapid iteration
 - Implemented **robust guardrails, failure handling** (PII filters, prompt constraints, rollback), reducing unsafe outputs by **50%**
 - Deployed and scaled via **FastAPI + observability stack** (Prometheus, OpenTelemetry) with p95 latency SLOs, drift detection
 - Iterated across prompts, retrieval strategies, and agent workflows, improving end-to-end answer quality under ambiguous requirements

- Excelerate — Data Analyst** **Aug 2023 – Sep 2023, Remote, USA**
- Led distributed team of 4 across time zones, managing execution using structured workflows and data-driven prioritization
 - Built BI dashboards (Power BI, Power Query, DAX) integrating data sources for real-time KPI tracking
 - Performed data cleaning and analysis (Excel, formulas, pivot tables), improving decision efficiency by **30%**
 - Delivered stakeholder reports, reducing analysis turnaround time by **25%** and improving decision clarity

- LetsGrowMore (LGMVIP) — ML Engineer** **Jun 2023 – Jul 2023, Chennai, IN**
- Built end-to-end ML pipelines across NLP, time-series, and vision using Python and Scikit-learn
 - Developed LSTM stock prediction and RNN text generation models, improving sequence prediction accuracy by **15–20%**
 - Implemented CNN models for image classification and emotion detection, achieving **90%+ accuracy** on validation datasets
 - Rapidly shipped 5+ ML projects, iterating across models and improving performance through tuning and evaluation

Projects

- Vision-Language Model Fine-Tuning (Qwen2.5-VL-7B)** **Feb 2026 – Mar 2026**
- Fine-tuned VLM using **Unsloth + LoRA** ($r=16$, $\alpha=32$) for structured image-to-text generation with consistent schema outputs
 - Identified dataset bias impacting generation; trained on **T4 (14.7GB VRAM)** with **7.6GB peak memory**, **5.5 min training**, validating efficient low-rank adaptation

- Autonomous Research Agent with LLMs** **Aug 2025 – Sep 2025**
- Built full-stack AI research agent (FastAPI, Tailwind, Gemini), reducing summarization effort by **70%**
 - Implemented observability and scalable agent framework (lmnr, Playwright), reducing failures **40%**, improving usability **95%**

- Multi-Agent RAG with LLMs** **Jul 2025 – Aug 2025**
- Built multi-agent RAG system (5+ agents, tool-use), improving task efficiency **30%**
 - Engineered LLM + retrieval pipelines (ReAct, memory), improving accuracy **25%**, reducing latency **40%**

Leadership & Activities

- AI Technical Content Creator — LinkedIn** **Apr 2025 – Present**
- Built AI technical content presence, generating **50K+ impressions**, **26K+ reach**, and **600+ engagements** across posts.
- Head of Events — WHHC & ACE Club, SRM — Event Lead** **Aug 2022–Sep 2023**
- Spearheaded hackathons and design challenges with **1,000+ participants**; ensured accessibility for all attendees.

Skills

Agent Design Patterns: ReAct, Plan-and-Execute, Reflection, Self-Refine, Multi-Agent Collaboration, Tool Use, Structured Outputs, Vector Databases, Embeddings

Agent Infrastructure: LangGraph, MCP, Context Engineering, Workflow Orchestration, Agent Memory, State Machines

Retrieval & Knowledge Systems: Agentic RAG, LlamaIndex, Hybrid Search, Query Routing, Reranking, Knowledge Graphs

Evaluation & Reliability: Evals, Harness Engineering, LangSmith, Phoenix, Guardrails, Tracing, Observability, Monitoring

Certifications: DeepLearning.ai TensorFlow Specialization; Neural Networks & Deep Learning

Publications: Brain Tumor Classification with Transfer Learning (ResNet-50, IEEE 2023)